Huawei Access Node BAS-ACC-28 Optical Transport Failure Resolution SOP

Node: BAS-ACC-28 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9304

Description

The Huawei Access Node BAS-ACC-28 is a critical component of the Basavanagudi POP, providing high-speed internet access to numerous customers. The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) failure domain on this node is a result of the intricate interplay between the hardware and software components of the device. Specifically, the failure is attributed to the degradation of the SFP/XFP transceiver modules, which are responsible for transmitting and receiving optical signals over the fiber optic cables. When the SFP/XFP modules deteriorate, the received power levels (Rx) decrease, causing the node to detect a fault and generate the ALARM_9304 alarm. Moreover, fiber cuts or Rx/Tx power loss can also cause the node to malfunction, resulting in the Optical Transport Failure failure domain. It is essential to identify and address the root cause of the failure to prevent data loss and service degradation.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and the node BAS-ACC-28 is physically available.

- Check the redundant link status to ensure it is operational and can take over in case of a failure.

- Review the node's alarm history to determine the time of the failure and any previous alarms.

- Gather information on the node's configuration, including the SFP/XFP module settings and fiber optic cable connections.

Potential Root Causes

Cause	Probability
SFP/XFP module degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault, Expected Output: Fault information, including fault ID, fault type, and fault location., Description: This command displays the fault information, allowing the technician to identify the specific fault type and location. The output will indicate whether the fault is related to the SFP/XFP module, fiber, or another component.

Command: display sfp, Expected Output: SFP/XFP module information, including status, temperature, and receive/transmit power levels.,
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Description: This command displays the SFP/XFP module information, allowing the technician to assess the module's status, temperature, and power levels. The output will indicate whether the module is functioning within normal parameters or if there are any issues.

Command: display fiber status, Expected Output: Fiber optic cable status, including fiber type, connection status, and transmission power levels., Description: This command displays the fiber optic cable status, allowing the technician to assess the fiber's type, connection status, and transmission power levels. The output will indicate whether the fiber is functioning within normal parameters or if there are any issues.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace sfp /1, Description: The technician will replace the degraded SFP/XFP module with a new one, ensuring that the module is properly seated and secured. After replacement, the technician will verify that the module is functioning correctly and the fault has been cleared.

Step 2: Step: Repair or replace the fiber optic cable, Command: replace fiber /1, Description: The technician will repair or replace the damaged fiber optic cable, ensuring that the connection is secure and the fiber is functioning correctly. After repair or replacement, the technician will verify that the fiber is functioning correctly and the fault has been cleared.

Step 3: Step: Adjust the SFP/XFP module settings, Command: set sfp /1 rx-power-threshold 0, Description: The technician will adjust the SFP/XFP module settings to ensure that the receive power threshold is set correctly. This will prevent the node from detecting a fault due to receive power loss.

Post-Check Verification

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Command: display fault, Expected Output: No faults detected., Description: This command verifies that the fault has been cleared and no additional faults are present.

Command: display sfp, Expected Output: SFP/XFP module information, indicating proper operation., Description: This command verifies that the SFP/XFP module is functioning correctly and within normal parameters.

Command: display fiber status, Expected Output: Fiber optic cable status, indicating proper operation., Description: This command verifies that the fiber optic cable is functioning correctly and within normal parameters.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore sfp /1, Description: If the replacement module is not available or the replacement fails, the technician will restore the original SFP/XFP module to ensure service continuity.

- Step: Restore the original fiber optic cable, Command: restore fiber /1, Description: If the repair or replacement of the fiber optic cable fails, the technician will restore the original fiber to ensure service continuity.